

Turning the **Hanky** Into a Smartphone



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This is not a 'frog turned into a prince' story, but a very long quest by technologists.

Technologists and, yes, advertisers, started dreaming about printed and flexible electronics decades ago, as early as 1970s. Products they wanted to develop evolved over time, but the core requirement of such printed electronics remained the key. The quest for large and flexible banners that could deliver interactive ads, and tablecloths that could be computer displays, evolved into the search for phones that could be folded up like handkerchiefs and flexible digital watches that would cling to your wrists. This continues to grow into more complex goals like flexible body implants and printable human tissues.

What makes today different from 1970s, however, is that dreams are seasoned with reality now. With several commercial products, demonstrations and promising researches, it is evident that printed, flexible and even organic electronics are fast becoming a rampant reality.

Research in this area predominantly classifies under manufacturing methods and processes, both lab-scale and large-scale,

materials for substrates and inks, as well as designs. In this story, let us take a look at some such interesting projects transpiring at universities and other research and development (R&D) centres across the world.

Printed spacecrafts and atmospheric confetti

How small would space centres become if spacecrafts could be printed? Well, we would know soon enough. NASA's Jet Propulsion Laboratory (JPL) has been exploring the idea of designing and fabricating a spacecraft entirely with flexible substrate printed electronics. The idea is to print a two-dimensional sheet with all the functional subsystems of a typical spacecraft, right from measurements to communications.

At the end of the project's first phase, the team reports, "Atmospheric confetti. Inchworm crawlers. Blankets of ground-penetrating radars. These are some of the unique mission concepts that could be enabled by a printable spacecraft." The low mass, volume and cost of printed spacecrafts makes NASA believe that, network missions would transform from a few discrete measurements to millions of platforms, achieving greater areal density and system reliability. Printed platforms could be released not just into space but also into volcanic plumes to measure composition and impact energies. These could be fitted in smart solar sails or areal balloons to monitor erstwhile unreachable areas and parameters.

The final report of the project's second phase affirms that, it is feasible to build an entire spacecraft out of printed electronics, but it raises doubts about launching such a spacecraft off a rocket and having it orient and propel itself to the intended destination. In the near term, applications might be simpler, such as sensors.

Printing power, literally

Last year, California based Imprint Energy demonstrated the print manufacturing of

Leaf-shaped, decorative, organic solar panel developed by VTT, Finland. Each leaf has an active surface of 0.0144m² and includes connections and a decorative part

